

VARUN KANNEPALLI

☎ 240-474-3677 | ✉ vkannepa@terpmail.umd.edu | 🔗 [linkedin.com/in/varun-kannepalli-b68a25396](https://www.linkedin.com/in/varun-kannepalli-b68a25396) | 📍 North Potomac, MD

EDUCATION

University of Maryland

B.S. in Mechanical Engineering

– **Minor:** Robotics and Autonomous Systems (RAS)

College Park, MD

May 2028

Wootton High School

High School Diploma

Rockville, MD

May 2024

EXPERIENCE

Clarkson University (AIR Lab)

Sensor Development and Implementation Pipeline Research Intern

Potsdam, NY

Jun. 2026 – Aug. 2026

- Designed and 3-D printed a rotatable sensor mount to track 360° spatial particle profiles across curved pipe geometries.
- Evaluated size-dependent particle, fluid momentum, and centrifugal dynamics under various flow conditions.
- Presented my findings at Clarkson University's Research and Project Showcase (RAPS).

NASA

Geospatial Analyst

Greenbelt, MD

Jun. 2022 – Aug. 2022

- Developed algorithms for geospatial data analysis, processing, and interactive spatial data visualization.
- Utilized advanced Geographic Information System (GIS) software tools to significantly enhance data precision.
- Created detailed reports and technical presentations delivered to agency stakeholders.

PROJECTS

Truss Design Project

- Constructed a wooden truss using pine wood and Gorilla Glue engineered to support over 1,000 lbs under strict size and weight constraints.
- Conducted structural analysis assessing internal forces, buckling modes, and safety factors, achieving a predicted failure load of 1,318 lbs and a strength-to-weight ratio of 4,903.

Beam Design Project

- Engineered a high-efficiency pine I-beam using MATLAB, Excel, and CAD to optimize dimensions surpassing a 1,000 lbs constraint.
- Evaluated bending failure modes and structural safety factors, predicting a 2,300 lbs failure threshold and achieving a 4,440 strength-to-weight ratio.

LEADERSHIP & ACTIVITIES

Terps Racing (FSAE EV)

Member

College Park, MD

2026 – Present

Qubo Robosub team (Robotics @ Maryland)

Member

College Park, MD

2026 – Present

UMD Anokha (South Asian A Cappella)

Arranger & Soloist

College Park, MD

2024 – Present

- Arranger and soloist for University of Maryland's South Asian A capella team.
- Placed 2nd Overall at the 2024-25 All American Awaaz (A3) Competition.

First Lego League

Competitor and Mentor

Gaithersburg, MD

2017 – 2023

- Competed for 4 FLL seasons, earning the Champions Award, Best Robot Design Award (2018), and Global Innovation Award (2019).
- Mentored FLL team "CCC Builders" on mechanical design and match strategy, leading to a 1st-place finish at the 2023 Bowie Qualifier.

Carnatic Vocal Performance

Duo Performer

DC Metro Area & India

Present

- Perform Indian classical (Carnatic) music concerts along with my brother.
- Performing 10 concerts during this year's December (Margazhi) Music Festival.

SKILLS

- **CAD:** SolidWorks, Autodesk Fusion 360
- **Coding & Software:** Python, Java, C++, MATLAB
- **Engineering Fundamentals:** Structural Analysis, Finite Element Methods, Fluid Dynamics
- **Manufacturing tools :** Lathe, 3D printer